



**BALL STATE**  
**UNIVERSITY**

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# **Module 3 Lecture - Probability Topics**

Introduction to Statistical Methods

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# 1 Overview and Introduction

## 1.1 Textbook Learning Objectives

- Understand and use the terminology of probability.
- Determine whether two events are mutually exclusive and whether two events are independent.
- Calculate probabilities using the Addition Rules and Multiplication Rules.
- Construct and interpret Contingency Tables.
- Construct and interpret Venn Diagrams.
- Construct and interpret Tree Diagrams.

## 1.2 Instructor Learning Objectives

- Begin to appreciate how probability connects back to the broader idea of statistics
- Understand the emphasis of chance and probability, not certainty, in statistics

## 1.3 Introduction

- Most decisions we make are a (hopefully educated) \_\_\_\_\_.
  - E.g., Should I wear my boots if it might rain today?, What are the chances that my class will respond well to this activity?
  - Chance and probability is \_\_\_\_\_, and, if we introspect closely, it plays a role in pretty much everything we do



Discuss: Try coming up with more examples of weighing probability in your own life

- Probability is a core part of \_\_\_\_\_, and our results are often phrases in recognizing that level of uncertainty
  - This is why you never really hear good scientists say we have definitively \_\_\_\_\_ something - there's always room left for uncertainty
- There is a lot of \_\_\_\_\_ (and math) surrounding probability
  - You could take an entire semester long class just talking about probability and the \_\_\_\_\_ /math of it

- We'll try to stay centered on exploring it as relevant to statistics, though this module may initially feel very removed from statistics

**! Important**

Even though we won't be 'in the weeds' as much with probability calculations later on, it will remain an important background subject!

## 2 Terminology

### 2.1 Introduction

- Like with the last module, \_\_\_\_\_ makes perfect
  - Please use the various exercises in the book and visit GA and/or professor office hours!
  - This is also a great time to embrace flashcards, if that historically works well for you

### 2.2 Rapid Fire Vocabulary

- In the context of probability, an **experiment** is some \_\_\_\_\_ controlled and monitored action - similar to how we use the term in research design
  - If the experiment is not **pre-determined**, that is, it's outcome is not \_\_\_\_\_, then there is some amount of **chance**
  - So if we call something a \_\_\_\_\_ experiment, its some controlled action that we are not sure of the outcome
  - Classic example is a coin flip

**? Example of how probability is intuitive - what is the chance that I flip a coin and it lands on 'tails'**

- A) 25%
- B) 75%
- C) 50%
- D) 0%

Explanation:

- The final \_\_\_\_\_ of the experiment is called an **outcome**
  - The possibilities of \_\_\_\_\_ the different outcomes is called the **sample space** (notation =  $S$ )
  - Sample space is \_\_\_\_\_ as a *list*, a tree, or Venn diagram; more example of these later
  - Coin flip with possible heads ( $H$ )/tails ( $T$ ) outcome example =  $S = \{H, T\}$
- An **event** is some \_\_\_\_\_ of outcome(s)
  - Important to know that it could just be \_\_\_\_\_ outcome or more than one described outcome
  - \_\_\_\_\_ will be defined as upper case letters like  $A, B, C$ , etc.
- The **probability** of a certain \_\_\_\_\_ occurring is described as the **long-term relative frequency** of said event
  - What that means is that the probability describes how \_\_\_\_\_ a certain event would occur if the experiment were to be completed an infinite number of times; this hinges on the **law of large numbers**
  - The \_\_\_\_\_ of a certain event happening will be written as  $P(x)$ , where  $x$  is whatever event is being looked at
    - \*  $P(x)$  will always be  $\geq 0$  and  $\leq 1$
  - When  $P(x)$  of all possible \_\_\_\_\_ is equal to one another, we would say that this experiment is **equally likely**, sometimes also said to be **fair**
- Example of coin toss, where event  $A$  is the likelihood of flipping with an outcome of head on a single toss, and  $B$  is the likelihood of tails
  - If  $P(A) = 0.50$ , i.e., a 50% chance to land on heads
  - If  $P(B) = 0.50$ , i.e., a 50% chance to land on tails
  - If  $P(A) = P(B)$ , i.e., equally likely experiment

 Discuss: Try writing out the same set of bullets that I just did, but with a six-sided equally likely die instead. We will calculate it out in a second, but try to just do this intuitively now.

## 2.3 Calculating Probability

- In the scenario that we aim to calculate the probability of a single \_\_\_\_\_

in an equally likely \_\_\_\_\_ space:

- Steps to \_\_\_\_\_ :
  1. Determine all \_\_\_\_\_ outcomes from the experiment and list them out, counting the number of possible outcomes
  2. \_\_\_\_\_ a specific event
  3. Assess how \_\_\_\_\_ outcomes match that events
  4. \_\_\_\_\_ the number of outcomes that match the event by the total number of outcomes;  $P(x) = \text{Matching} / \text{Total}$
- Do a coin toss of a dime and nickel
  1. 4 outcomes in the sample space:  $\{HH, TH, HT, TT\}$  where
    - $H$  heads on a coin
    - $T$  tails on a coin
    - In writing this would be: both coins land on heads, dime lands tail and nickel lands heads, dime lands heads and nickel lands tails, both coins land on tails
  2. Event  $A$  of getting at only one heads; could also be written as  $A : n_H = 1$
  3. Two outcomes in the sample space match this criteria:  $/TH, HT/$
  4. Thus the probability of event  $A$  is calculated as  $P(A) = 2/4 = 0.50$

 Discuss: Trying doing the same full calculation process now, but now for a 6-sided equally likely die.

- Practically, many events are not \_\_\_\_\_ fair / equal - coin flips are affected by the person doing them, air resistance, etc.
  - This will be called **unfair or biased**

## 2.4 OR Events

- An **OR Event** is when an event is written in such a manner that an outcome could \_\_\_\_\_ event  $A$  or  $B$ 
  - This would be written as  $A \cup B$  (that symbol is called a cup)
  - Applying the above, we would have a \_\_\_\_\_ step process of determining the outcome match  $A$  OR  $B$  and would give the probability as  $P(A \cup B)$

- To list the possible outcomes, we find the **set union** of the two sets of outcomes that match the \_\_\_\_\_ events, example:
  - If  $A$  matches  $\{1, 2\}$  and
  - If  $B$  matches  $\{2, 3, 4\}$  then
  - $A \cup B$  matches  $\{1, 2, 3, 4\}$
  - Then we proceed by using  $A \cup B$  matches in step 4 of [Calculating Probability](#)

## 2.5 AND Events

- An **AND Event** is when an event is written in a manner that an outcome could match event  $A$  or  $B$ 
  - This would be written as  $A \cap B$  (this symbol is a cap, like a hat)
  - Thus probability would be written as  $P(A \cap B)$
- To list the possible outcomes, we find the **set intersection** of the two sets of outcomes that match the \_\_\_\_\_ events, example:
  - If  $A$  matches  $\{1, 2, 3\}$  and
  - If  $B$  matches  $\{2, 3, 4\}$  then
  - $A \cap B$  matches  $\{2, 3\}$
  - Then we proceed by using  $A \cap B$  matches in step 4 of [Calculating Probability](#)

### ! Important

A mnemonic to remember caps and cups: cAps are for Ands

## 2.6 Conditionals and Complements

- **Complements** take the \_\_\_\_\_, or everything but the matching outcomes of an event
  - This is written at the \_\_\_\_\_ of an outcome, e.g.,  $A'$  is the complement of  $A$
  - Thus, probability of a conditional would be notated as  $P(x')$
  - In such a scenario, instead of finding all the outcomes that DO match the event, we would list all the outcomes that \_\_\_\_\_ match the event in step 3 of [Calculating Probability](#)
- **Conditionals** are the probability a certain event occurs \_\_\_\_\_ that another event has already happened
  - It is written as  $P(A|B)$ , which translates to probability of  $A$  given  $B$  (the vertical bar is sometimes called a pipe)
  - This is calculated via the following formula, assuming  $P(B) \neq 0$ :

$$P(A|B) = \frac{A \cap B}{B}$$

- Once again, try working through the problems and solutions in your book for all of these scenarios!

### 3 Independent and Mutually Exclusive Events

#### 3.1 Introduction

- We are often interested in how two possible events are \_\_\_\_\_ related or unrelated
  - We can investigate both the independence and mutually exclusivity of events to determine this
  - These are both \_\_\_\_\_ description

#### 3.2 Independence

- Two events are considered to be **independent** if they are unrelated/do not \_\_\_\_\_ one another
  - On the contrary, if they are somehow \_\_\_\_\_ to one another, they are said to be **dependent**
- The following are the \_\_\_\_\_ rules to show independence:
  - $P(A|B) = P(A)$
  - $P(B|A) = P(B)$
  - $P(A \cap B) = P(A)P(B)$  (The “Multiplication Rule”)
  - Only one of these conditions needs to be met for establishing independence
- It is safest to \_\_\_\_\_ dependence of events until independence has been established
  - Sort of counter-intuitive!

##### 3.2.1 Connection to Sampling Process

- Recall that in a *random* sampling scenario, there is a theoretically \_\_\_\_\_ chance for each member of the population of interest to be selected
  - Effectively, random sampling is sampling that is done *equally likely*
- Sampling can be done one of two ways, that impact whether sampling decisions were made \_\_\_\_\_ of one another

- Sampling can be done **with replacement** which means the same individual could be chosen \_\_\_\_\_ times - this means the odds of being selected as a member of the population remain the same
- Sampling can also be done **without replacement**, meaning it is done in a manner that a person \_\_\_\_\_ be chosen multiple times, thus reducing the outcomes by one and changing the odds for each subsequent selection

### 3.3 Mutually Exclusivity

- Events are **mutually exclusive** when they cannot occur at the same time and follow the “Addition Rule”:

$$P(A \cap B) = 0.00$$

- If unsure, assume \_\_\_\_\_ until this rule is met

## 4 Contingency Tables

### 4.1 Introduction

- The word “contingent” means to depend on, so **contingency tables** offer a nice view for displaying probabilities in such a way to \_\_\_\_\_ dependence

#### ! Important

Like with descriptive statistics, using graphical and tabular displays can make it much easier to navigate complex information, make sure you use them!

## 4.2 Example of a Contingency Table

|                                          | Speeding violation<br>in the last year | No speeding violation<br>in the last year | Total |
|------------------------------------------|----------------------------------------|-------------------------------------------|-------|
| Uses cell phone<br>while driving         | 25                                     | 280                                       | 305   |
| Does not use cell<br>phone while driving | 45                                     | 405                                       | 450   |
| Total                                    | 70                                     | 685                                       | 755   |

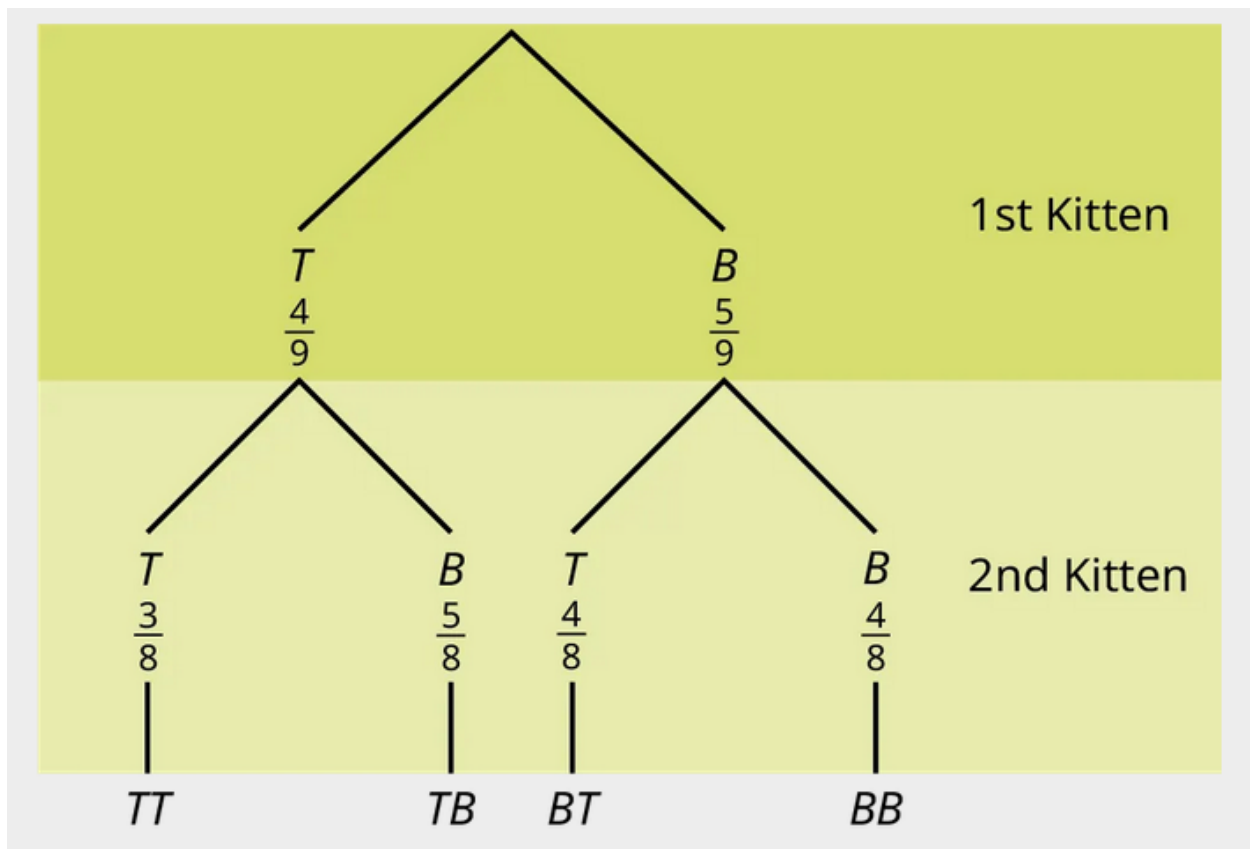
Table 3.3

## 5 Tree and Venn Diagrams

### 5.1 Introduction

- Just like with how descriptive statistics have several ways to graphically present, so do probabilities!
  - Contingency tables are a great start, but we can add to them with \_\_\_\_\_ diagrams and \_\_\_\_\_ diagrams
- For this section, I'll be a bit more brief only because these representations are not super \_\_\_\_\_ in most applied statistics

## 5.2 Tree Diagrams



## 5.3 Venn Diagrams

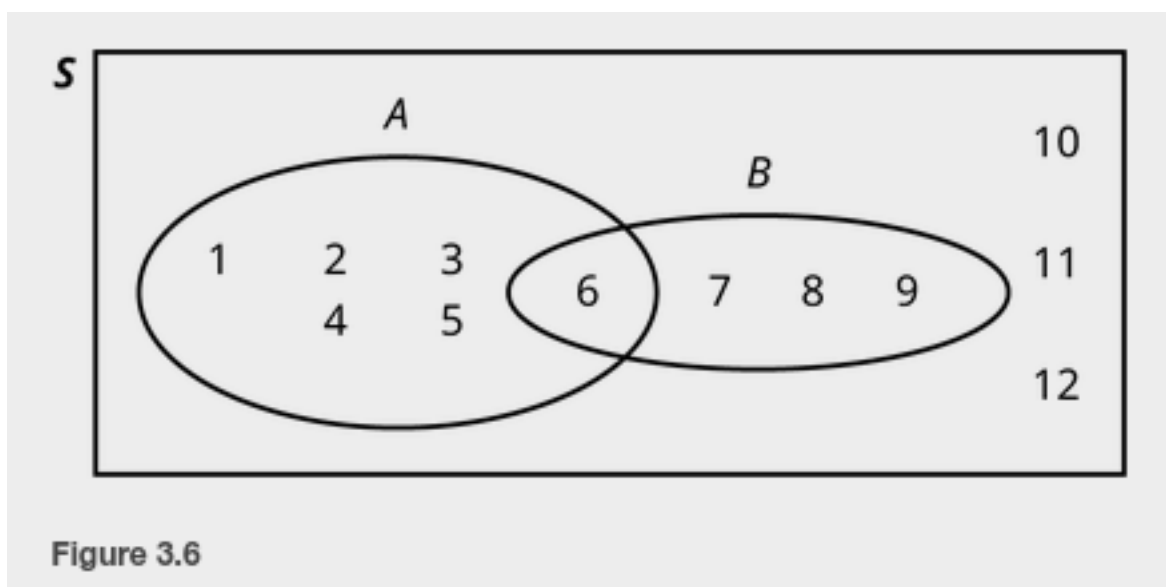


Figure 3.6

## 6 Conclusion

### 6.1 Recap

- Probability is a tough subject, but it is critical as a foundation topic prior to understanding inferential statistics!
- Understand that when we discuss statistical results going forward, it will always be in through the lens of probabilities and likelihood of outcomes.
- While “real” statistics won’t often involve intensive hand calculation of probabilities, you should still have a foundation appreciation of how to do it - just like we did with standard deviation.

### 6.2 Lecture Check-in

- Make sure to complete and submit the lecture check-in
- Also make sure to attend to additional SPSS and practical assignment this week!